

Urban Wage Premium*

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Abstract

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*People to thank...

1. Introduction

As of 2015, workers in urban areas earn a 16 percent premium over their non-urban counterparts. The urban wage premium has become an empirical fact, widely documented in the urban economics literature (see for example Rosenthal and Strange (2004) for a review). The literature has closely explored the role of high productivity individuals sorting into cities (Combes et al., 2008). For example Yankow (2006) documents that up to two-thirds of the urban wage premium is driven by differences in worker skills. The literature has also closely studied the role of differential returns to experience in urban areas as a source of the wage premium (Baum-Snow and Pavan, 2012). Outside of human capital explanations, researchers have explored the role of agglomeration economies and firm survival to explain the remaining gap (Glaeser and Maré (2001), Wheaton and Lewis (2002), Combes et al. (2012), Behrens et al. (2014)).¹ Agglomeration economies are an attractive explanation, yet determining the underlying mechanism that contributes to agglomeration is difficult. Indeed a large literature has sought to explain the causes and mechanisms of agglomeration.

According to Marshall et al. (1920) physical proximity and connections to firms are a drivers of agglomeration. Similarly, Duranton and Puga (2004) and Head and Mayer (2004) stress how trade costs may shape agglomeration and industrial location, and thus may partially explain the observed urban wage premium. In this paper, we directly explore how improvements in transportation that shape connections between locations, measured by market access, impact the estimated urban wage premium in the United States between 1940 and 2015. Empirically we begin by estimating regres-

¹Recently, work has also shown that urban industry-level concentration has led to a markdown of wages as firms exercise monopsony power Azar et al. (2020), thus, in the absence of concentration, the urban wage premium may be even larger than previously thought.

sions of individual wages on a set of individual characteristics and location and time fixed effects. This set of regressions enables us to partial out the portion of the wage gap due to underlying differences in the skill-education-experience composition of workers between urban and non-urban areas, while at the same time, netting out common firm survival trends that contribute to the wage gap. Similar to prior work, we find that individual characteristics account for approximately half of the urban wage premium and become increasingly important after 1980. We then construct a measure of market access following Donaldson and Hornbeck (2016) and Jaworski and Kitchens (2019) at the MSA level drawing on newly digitized atlases of the US Highway Network each decade 1940-2010. Our measure of market access allows us to consider how changes in a city's connectedness to other locations affects the wage premium. In general, we find that market access (MA) has a direct, positive impacts on wages, but the market access variable is uncorrelated with the urban variable, thus it has little to no impact on the urban wage premium. Importantly, we find that a portion of market access is collinear with urbanization, thus it is important to measure market access, net of the own market effect. This should be clear as by definition market access captures local productivity, population density, and amenities. Combined, the MA estimates highlight the importance of localized productivity differences, that we explore in more detail.

In particular, we are interested in understanding how changes in MA, induced by expansions of the highway system have influenced each MSA's position in the wage premium distribution. That is, as the highways have expanded, has it altered the rank of specific cities by the size of the premium. This exercise relates to work by Yankow (2006), who explores heterogeneity in the wage premium among large and small MSAs.

Finally, we seek to understand to what extent spatial sorting on unobservables, such as unobserved amenities or unobserved productivity, drives

the urban wage premium. To address concerns regarding selection on unobservables, we follow Altonji and Mansfield (2018) and control for MSA level averages of individual characteristics in additional specifications. The inclusion of these MSA-level averages should, in theory, net out the influence of unobservable productivity, for example, due to higher returns to experience in urban areas, as described by Baum-Snow and Pavan (2012). For example, consider the unobservable trait of working longer hours, as discussed in Weinberg et al. (2004). If workers in a particular city work longer hours due to an unobserved and time varying preference for longer work days, then this may be reflected in the urban wage premium. Controlling for the MSA-level averages nets out this type of sorting, however it may overcontrol by limiting the scope for other mechanisms. In this case, controlling for observables to address unobserved preferences for longer hours may limit the scope of referrals from neighbors, as discussed in Bayer et al. (2008). In addition to controlling for MSA-level averages of individual characteristics, we also estimate specifications where we control for MSA level housing values. We add housing value controls to limit the impact that unobserved amenities or land use restrictions have on wages.

When we add MSA-level averages of individual characteristics and MSA average housing values to our specifications, we find that the urban wage premium rises/falls by XYZ. Overall, our findings highlight the role that selection, both by individuals and firms, contributes to the urban wage premium. Indeed, after controlling for many sources of selection, we find that the unexplained wage premium is XX percent of the observed premium.

In what follows, we provide a brief background of why cities formed and how transportation has evolved over time. We then describe our data and empirical strategy.

2. Historical Background

2.1 The Rise of Cities

As of 2019 over 82 percent of Americans live in urban areas, up from just under 40 percent in 1900. Many factors explain the rise and growth of cities throughout the 20th Century. First, both biological and mechanical innovations in agriculture altered patterns of crop specialization and the optimal mix of labor and capital, freeing both land and labor for alternate uses (Caselli and Coleman II (2001), Michaels et al. (2012), Olmstead and Rhode (2001), Griliches (1963), and Jung (2020)). Second, negative health externalities due to crowding in cities has been largely mitigated through public health efforts. Early in US history, cities were dirty, unhealthy places that required a wage premium to attract workers to face risks associated with infectious diseases such as cholera, typhoid, yellow fever, and smallpox (Cutler and Miller (2005), Beach et al. (2016), Ferrie and Troesken (2008), Alsan and Goldin (2019), and Anderson et al. (2020)). As urban health improved, the need to attract workers with high wages subsided, however, as population density increased, so too did rents. Finally, there have been drastic changes in transportation that have facilitated the growth of cities.

Cities initially formed in places where it was convenient to trade and process agricultural goods due to natural advantages (Boustan et al., 2013). With improvements in technology, the scale and scope of cities has changed. For example, new cities sprang up near navigable rivers and canals to support the shipment and transshipment of goods. Later, the railroads bypassed many previous cities and led to new urban centers.² In the post-WWII era, there have been significant improvements in car and truck transportation that have further expanded and reshaped cities. For example, Rephann and

²As Bleakley and Lin (2012) demonstrate, agglomeration forces allow cities to exist after their technological obsolescence.

Isserman (1994) and Duranton and Turner (2012) highlight the role of the interstate highways in expanding growth rates and city-level employment in the United States. In the context of shipping transportation improvements, Brooks et al. (2018) and Ducruet et al. (2019) highlight how population has expanded as a result of containerization, typically in land abundant cities. Outside of the United States, work by Faber (2014) and Baum-Snow et al. (2020) document that trunk highway systems in China shifted activity from rural to urban areas. While the existing literature has shown that cities grow as a result of better transportation, it is less clear whether the growth is accompanied by an increase in the urban wage premium, due for instance to new agglomeration economies.

In our empirical exercise below, we will test whether improvements in market access stemming from internal highway improvements contributes to both wage growth and the urban wage premium.

2.2 Highway Background

In 1940, there were 3.28 million miles of road in the United States.³ Our data suggests that there were just over 358 thousand miles of federal and state aid highways in 1940. Of these, 126 thousand miles of US Highway were paved, 8.8 thousand miles were improved, but not paved, and over 2 thousand miles were dirt roads.⁴ The remaining 220 thousand miles of road in 1940 were state highways, of which just over 60 percent was paved.

Since 1940, federal and state highway networks have improved dramatically. First, over 1 million new miles of road were added to the network, although much of this mileage is maintained by local governments and is

³<https://www.fhwa.dot.gov/policyinformation/statistics/2018/vmt421c.cfm>

⁴Our measure of mileage may differ from other measures given our data procedure used to develop time harmonized highway networks. For example, rather than deleting segments and recoding segments over time, which generates 126,000 miles of paved US Highway, tracing 1940 paved US Highways results in 127 thousand miles.

for local travel.⁵ Of the 1 million new miles, approximately 50 thousand were built or upgraded to create the Interstate Highway System (IHS), which drastically increased travel speeds between places in the network. As highlighted in Michaels (2008), the IHS construction was especially active in the 1960s and 1970s. In addition to the construction of the IHS, other upgrades to state and non-IHS Federal-Aid Highways occurred.⁶ These smaller improvements include straightening road alignments, upgrading road surfaces, road widening, and urban bypasses; all of which increase travel speeds. In the post WWII era, there has also been an effort to use road construction as a local development tool. Today there are over 220,000 miles in the National Highway System, reflecting expansions and improvements of both the IHS and the US Highway network.⁷

The improvements to the highway network have had meaningful impacts on travel times in the United States. In Figure 1, we highlight how the average travel time between all county pairs (4.6 million unique routes) in the United States has changed between 1940 and 2010.

In 1940, the average travel time was 27.67 hours. By 2010, the average travel time had fallen by 9.44 hours, to 18.23 hours, a 34 percent relative decline.⁸ The largest drops in travel time coincide with the completion of major portions of the IHS occurring between 1960 and 1970. As we document below, the changes in travel time varied significantly across space. In part, this is due to the funding rules of the IHS, which disproportionately favored the South and West.

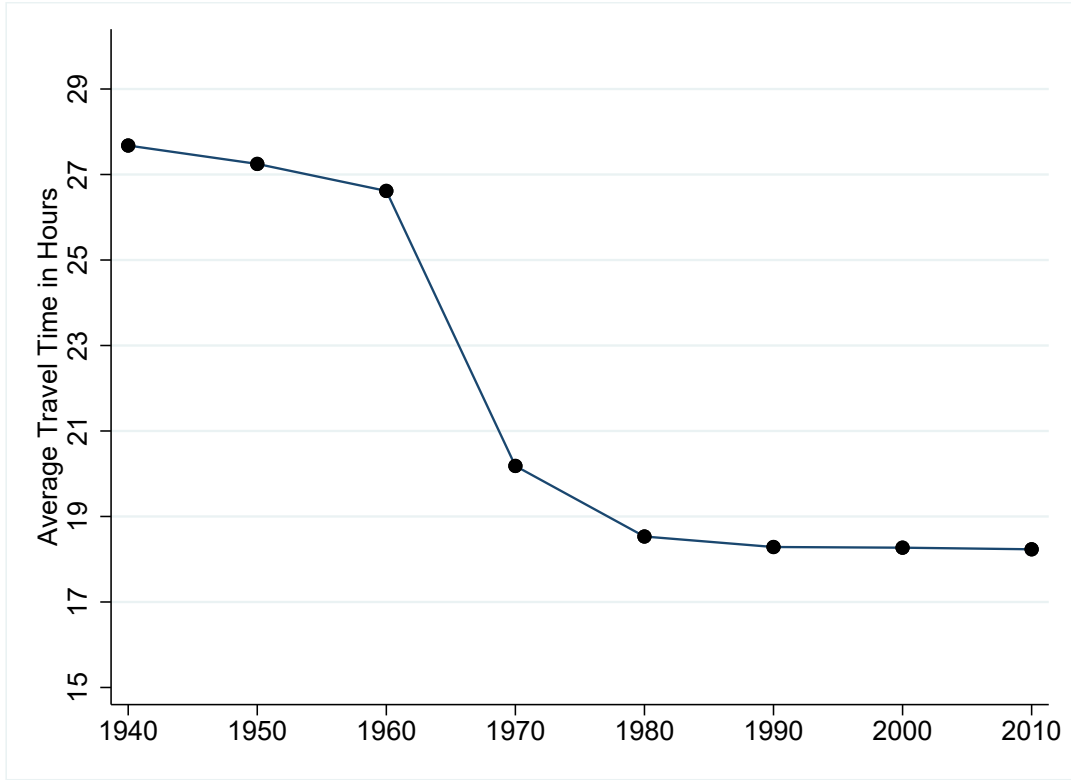
⁵<https://www.fhwa.dot.gov/policyinformation/statistics/2018/vmt421c.cfm>

⁶Examples of non IHS improvements include the construction of the Appalachian Development Highway System, improvements to the Strategic Highway Network (STRAHNET), the Florida Turnpike, Texas Farm to Market (FM) Roads, among many others

⁷<https://www.fhwa.dot.gov/policyinformation/statistics/2018/hm14.cfm>

⁸While travel times fell, the average route increased by 13 miles, or just over 1 percent.

Figure 1: Average Travel Time in Hours 1940-2010



Notes:

3. Data

To estimate the relationship between the urban wage premium and the role that market access has on the premium throughout the post war era, we draw on several datasets. First, we draw on publicly available microdata of decennial Censuses and the American Community Survey (ACS). We combine this individual level data with measures of market access constructed from a recently harmonized decadal dataset of the domestic US Highway Network that we describe in more detail below.

3.1 Wage and Individual Characteristic Data

To measure the average earnings per worker (wages), we use the IPUMS Full Count Census for 1940, the 1 percent sample Census for 1950 and 1970, the 5 percent sample Census for 1960, 1980, 1990, and 2000, and the 2005, 2010, and 2015 American Consumer Surveys. Our sample consists of adult men living in the continental US between the ages of 25 and 65 who are employed for at least 26 weeks in the sample year. We apply IPUMS person weights in our summary statistics and all our regressions.

We compute average weekly earnings (wages) by finding total income divided by the number of weeks worked in the given survey year. For individuals earning the top income code, we set their total income to the lower top income cutoff multiplied by 1.5.⁹ If the urban wage premium is primarily driven by people in the top income group, our estimates of the urban wage premium will be biased downwards by this choice. All incomes are then multiplied by the Consumer Price Index to be in 2015 dollars. In order to have a consistent measure of occupation, we use the ‘occ1950’ variable provided by IPUMS which harmonizes occupation codes across all our survey years. In Table 1 we report the number of individual observations in each cross section.

Table 1 also reports the average weekly urban and nonurban wage in each time period. Two striking trends appear from the summary statistics. First, as we noted previously, the share of the population living in an urban areas has steadily increased between 1940 and 2015. This trend matches those highlighted in Boustan et al. (2013). Secondly, it is clear that both urban and nonurban mean weekly wages are growing over time in 2015 \$ from \$33.24 and \$25.23 in 1940 to \$1353.95 and \$1067.48 in 2015. The urban wage pre-

⁹These cutoffs are reported in IPUMS INCWAGE documentation. In 2015 \$, the cutoffs are \$84,500 for 1940, \$99,200 for 1950, \$199,500 for 1960, \$309,000 for 1970, and \$225,000 for 1980. Afterwards, IPUMS reports income as the median wage above that year’s cutoff.

mium, the ratio of the urban to nonurban wage, follows a W shape. The urban premium starts at 0.32, and then falls sharply between 1940 and 1950, where it declines to a low of 19 percent in 1980. Since 1980, the urban wage premium has increased; in 2015 the premium was 26 percent.

Table 1: Summary Statistics

Year	N	% Urban	log(Market Access)		Urban Weekly Wage (2015 \$)		Nonurban Weekly Wage (2015 \$)	
			Mean	SD	Mean	SD	Mean	SD
1940	16,939,585	0.66	3.39	0.87	33	20	25	17
1950	68,457	0.69	3.38	0.67	72	41	59	33
1960	1,364,970	0.66	3.33	0.66	124	77	100	60
1970	280,338	0.75	3.42	0.70	205	138	166	105
1980	1,910,440	0.73	3.22	0.57	409	284	341	221
1990	2,262,149	0.74	3.12	0.46	701	562	558	405
2000	2,583,088	0.79	3.08	0.42	1,008	1,015	767	677
2005	530,737	0.79	3.08	0.42	1,143	1,146	863	712
2010	530,783	0.80	3.01	0.51	1,228	1,210	961	819
2015	561,934	0.81	3.00	0.52	1,353	1,446	1,068	1,011

Notes: Summary statistics weights individuals by their person weight. Weekly wage is reported in 2015 \$. Market access is calculated from solving $MA_c = \rho \sum_d \tau_{cd}^{-\theta} MA_d^{-1} Y_d$, where transportation costs, τ_{cd} , are calculated as described by Equation 1.

3.2 Market Access Data

To construct our measure of market access, we begin by drawing on recently digitized data by Jaworski and Kitchens of historic highways. For the years 2000, 2005, and 2015, we rely on the detailed shapefiles maintained by the

federal government. For years prior to 2000, we georeference atlases produced by Rand McNally in each decade 1940-1990, as the federal government did not begin maintaining its database until 1994. To create a time consistent network, we begin with the 2000 version of the Federal Highway Administration’s shapefiles. Then, for each decade, we overlay scans of each decade’s Rand McNally atlas and edit the federal shapefile. Specifically, we delete road segments that were constructed between the current map year and 2000, and re-code segments that have been reclassified over time (i.e., segments of the US Highway network that became interstates, or paved roads that were previously gravel or dirt). We perform this procedure iteratively for each decade back to 1940. One advantage of this approach is that it locates roads based on modern GPS technologies to ensure accurate placement throughout time.¹⁰ We then follow the process outlined in Jaworski and Kitchens (2019) to measure the distance and travel time through the network each decade for all origin-destination pairs. Specifically, we represent each location by its geographic centroid, connected to the network by a set of access roads. We assign a travel speed to each segment in the network according to its classification (state, federal, interstate) and surface (dirt, gravel, paved).¹¹ We then solve for the least cost route through the network for all pairwise combinations of origins and destinations.

The measures of distance and travel time are then used as inputs to mea-

¹⁰One drawback is that this approach may ignore roads that were important prior to the modern stock of infrastructure.

¹¹The speed limits used in this paper are Interstate - 65; US Highway (Paved) - 45; US Highway (Gravel) - 35; US Highway (Dirt) - 30; State Highway (Paved) - 25; State Highway (Gravel) - 20; State Highway (Dirt) - 15; Access Road - 10. Jaworski and Kitchens (2019) show that the measure of market access is robust to other speed limits.

sure the origin-destination trade costs, τ_{cd} . Formally,

$$\begin{aligned} \tau_{cdt} = & \text{distance in miles}_{cdt} \times \text{cost per mile}_t \\ & + \text{travel time in hours}_{cdt} \times \text{hourly wage of truck drivers}_t \end{aligned} \quad (1)$$

where the cost per mile is based on the cost per gallon and the number of miles traveled per gallon and we the hourly wage for a truck driver.

To construct a measure of market access, we follow the inter-regional trade literature developed in Eaton and Kortum (2002), Redding and Venables (2004), and Donaldson and Hornbeck (2016).¹² A county c 's market access, MA_c , is given by this system of non-linear equations

$$\text{MA}_c = \rho \sum_d \tau_{cd}^{-\theta} \text{MA}_d^{-1} Y_d, \quad (2)$$

where θ is the trade elasticity with respect to trade costs, ρ is a constant from the theoretical model, and Y_d is the income of county d measured by GDP.

The market access variable summarizes a county c 's access to the transportation network by taking an average of other counties' market size weighted by both the trade cost to ship to that market and by all other counties access to that market (MA_d^{-1}). Therefore a county c 's market access is large when it has access to large markets that other counties have little access to.

Since our measure of market access is increasing in a county's own size, there could be concern that our measure is not only captures the county's access to the transportation network, but also the size of the county itself. D'Costa and Overman (2014) finds evidence that the urban wage premium indeed is growing in city size. This would mean market access is absorbing a portion of the premium associated with being in a largely populated area and would attenuate our estimates of the wage premium towards zero. To

¹²The particular trade model used is described in Jaworski and Kitchens (2019).

avoid this, for all counties in an MSA, we construct a measure of market access that removes all counties from the summation that fall within the same MSA. This is similar to one of the approaches used by Donaldson and Hornbeck (2016) that omits counties within a given radius of the origin county. The equation we use for market access is therefore:

$$\text{MA}_c = \rho \sum_{d \notin \text{MSA}(c)} \tau_{cd}^{-\theta} \text{MA}_d^{-1} Y_d, \quad (3)$$

To aggregate market access to Urban areas, we created a crosswalk of counties within MSAs. For each MSA, we average across the counties within the MSA.¹³ Our measure of non-urban market access for a given state is the average of counties in that state that do not fall within any MSA. Therefore, for each individual we have a measure of whether or not they live in an urban area and the market access of the urban/non-urban area they live in. This process is detailed in Appendix B.

Table 1, Column 4 highlights how market access has increased over time. However, the summary statistics mask significant heterogeneity across space. In Figure 2 we document how the log of market access has changed over time.

While market access has grown everywhere, the South and West have experienced relative gains, in part due to the construction of the IHS and migration patterns. In Figure ?? we show the within-year deviations of the log of market access from the yearly mean at the MSA-level and the state average of non-urban counties. The figure shows that over time, market access in non-urban areas has grown relative to urban areas, particularly in the southern and western United States. On the other hand, large cities in the 1940 had very large market access relative to the national average and this relative advantage declines over time from about 4 log points to about 2 log

¹³Our results are robust to weighting by population. (NEED TO CHECK)

points above the mean. The deviation in log market access from the national average highlights the variation in market access used in our regressions below.

4. Estimation and Results

We begin by computing the log ratio between the average urban weekly wage and the average non-urban weekly wage for each survey year (adjusted for inflation using the BLS' consumer price index) following Boustan et al. (2013). This ratio captures the raw, unadjusted urban wage premium that has been discussed in previous literature. The results of this exercise are presented in Figure 3.

The urban wage premium starts around 30% in 1940, decreases between 1950 and 1980 to just over 20% and then increases again to over 25% after 2000. This is consistent with results found in the literature.¹⁴

The unadjusted urban premium reflects a combination of differences in productivity, the endogenous selection of labor into urban areas, differential returns to experience and other potential factors, such as agglomeration economies. To better understand how these different channels impact the urban wage premium, we move to a regression framework. Specifically, we estimate a set of regressions for each survey year as follows:

$$\log(\text{Wage}_i) = \beta_0 + \beta_1 \text{Urban}_i + X_i \delta + \varepsilon_i. \quad (4)$$

Here Urban_i is an indicator variable that takes the value of 1 if an indi-

¹⁴Our results match the shape over time found in Figure 3 of Boustan et al. (2013). Relative to prior work, our unadjusted difference for the year 2000 is smaller in magnitude than the estimate of 32% from Glaeser and Maré (2001). Despite this difference, our data generate a similar pattern to others documented in the literature, such as the increase between 1980 and 2000 described in Baum-Snow and Pavan (2012).

vidual resides in an MSA and is zero otherwise. In the absence of controls, β_1 captures the unadjusted urban wage premium. To understand what is driving the urban wage premium, in the following subsections we slowly introduce sets of control variables into X_i and reestimate equation 4.

Figure 4 presents the point estimates of β_1 in the set of specifications as well as a dotted line representing the method used by Boustan et al. (2013). The line with square markers is Equation 4 with no controls. The primary thing to note is that the urban wage premium no longer significantly rebounds after 1980 which counters previous descriptive work that shows a recent resurgence of the wage premium. The reason for the difference is simple. The method in Boustan et al. (2013) estimates the following

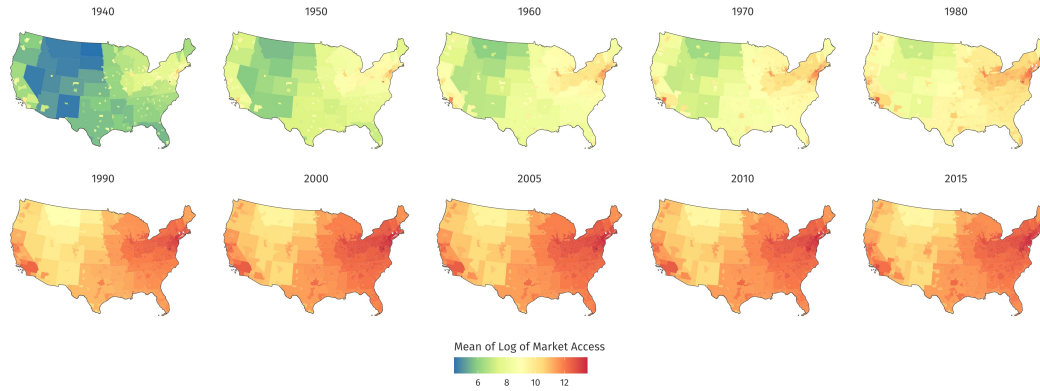
$$\log(\hat{\mathbb{E}}[\text{Wage}|\text{Urban} = 1]) - \log(\hat{\mathbb{E}}[\text{Wage}|\text{Urban} = 0])$$

which differs from the regression estimate

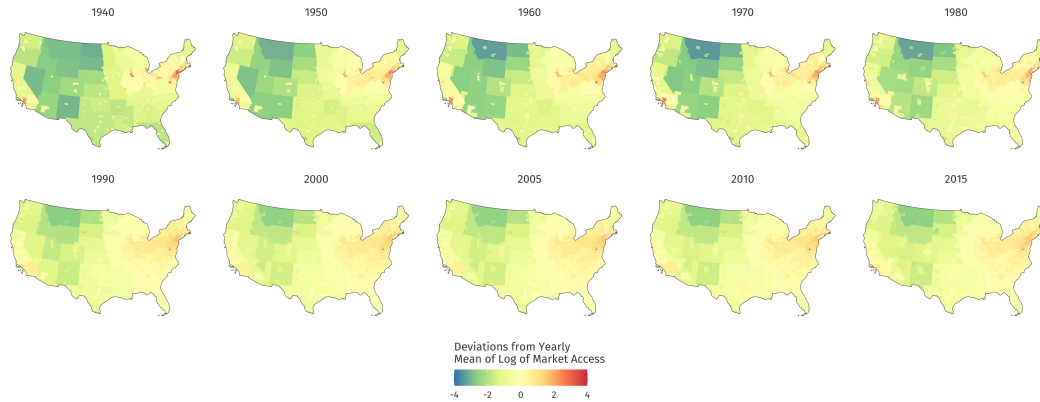
$$\hat{\beta}_1 = \hat{\mathbb{E}}[\log(\text{Wage})|\text{Urban} = 1] - \hat{\mathbb{E}}[\log(\text{Wage})|\text{Urban} = 0]$$

Figure 2: Market Access over 1940-2015

(a) Log of Market Access

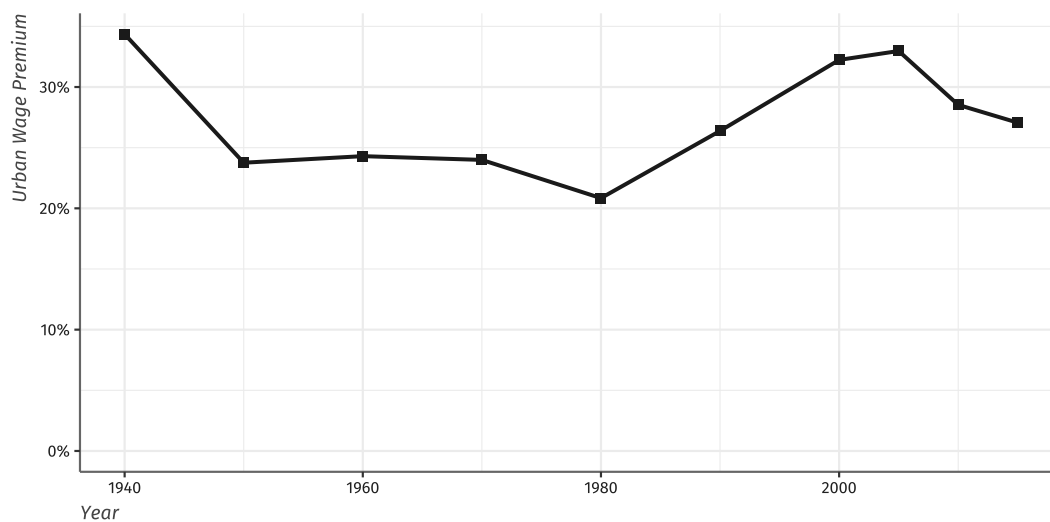


(b) Deviations from Yearly Mean



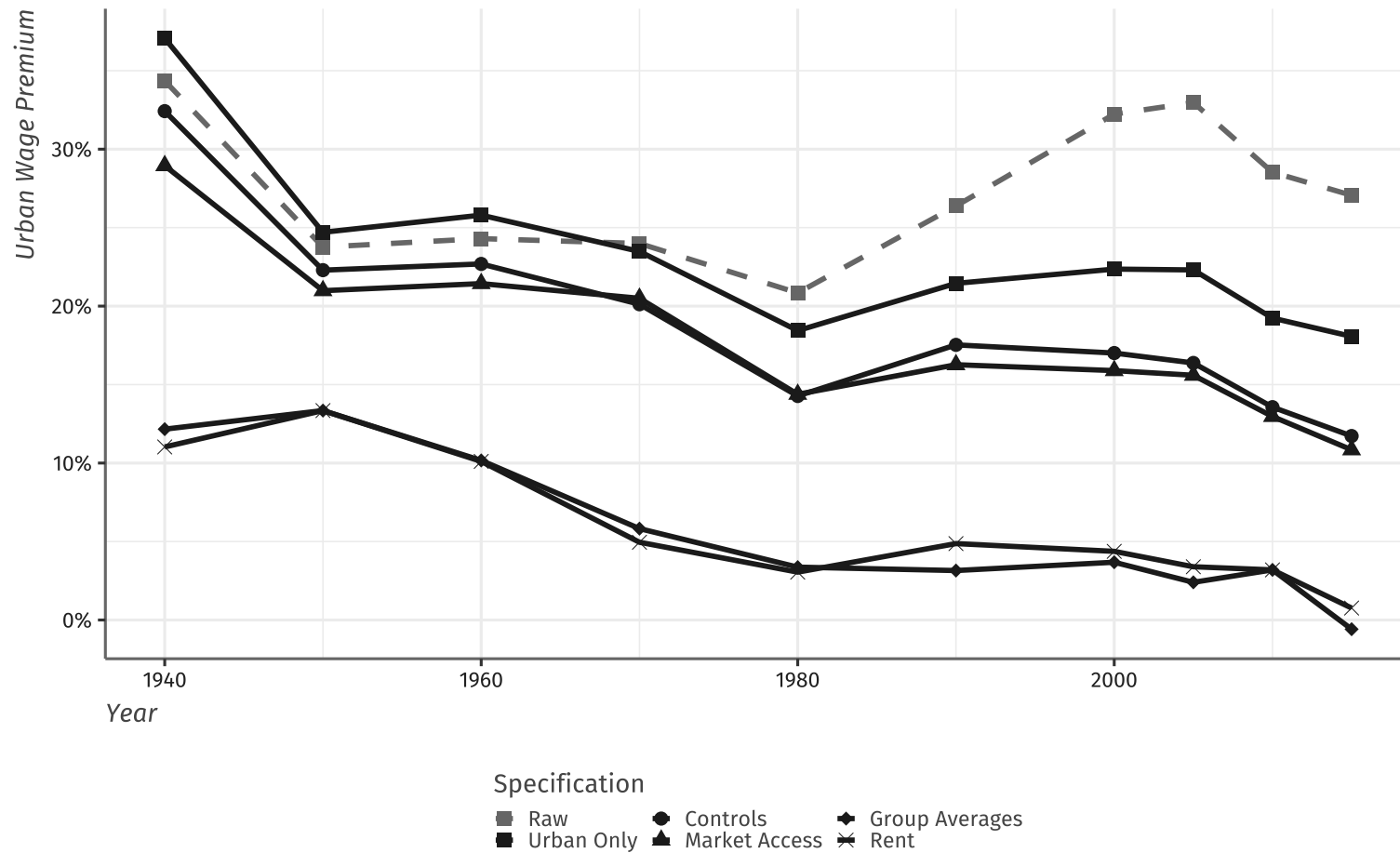
Notes: Panel (a) shows the log market access averaged to MSAs and the state-wide average of non-urban counties. Panel (b) shows deviations from the national mean of log market access averaged to MSAs and the state-wide average of non-urban counties. These averages are described in Appendix B. Market access is calculated from Equation 2 using trade costs calculated in Equation 1.

Figure 3: Urban Wage Premium 1940-2015 (Boustan et al., 2013)



Notes: Each point is the log of the ratio of average wage among urban male wage-earners and non-urban male wage-earners in the given survey year.

Figure 4: Urban Wage Premium 1940-2015



Notes: This figure plots the coefficient estimates for survey-year level estimates of the urban wage premium. In the grey line with square markers, the estimate is found by taking the ratio of the average log weekly wage for urban and nonurban workers. The remaining lines run versions of the regression given by Equation 4. In the line with diamond markers, there are no controls and no measure of market access; the line with triangle markers includes controls for age in 5-year bins, education levels, and an indicator for being White; the line with circle markers includes the same controls and the log of market access.

4.1 Individual Control Variables

There are many reasons to not interpret the bivariate regression coefficient as a causal parameter. First, people can freely sort between urban and non-urban areas and therefore the individuals may differ in their productivity and hence their wages. To limit the scope that endogenous sorting has on the urban wage premium, we control for a set of individual-level variables. Since we want to compare across years, we use the same set of individual-level covariates which limit the number of variables we can include to a set of 5-year age bins, indicator variables for education level, and an indicator for race.

We report the urban wage premium estimates after controlling for individual characteristics in Figure 2 by the line with circle markers. Our estimates suggest that in 1940, the individual-level controls explain about 1/8 of the unadjusted urban wage premium, but the shape of the premium over time remains consistent. However, the explanatory power of individual controls steadily grows over time. In 2015, the individual-level controls explain about 1/3 of the urban premium highlighting the importance of sorting of individuals in explaining wage premiums in the modern era. Across all years in the sample, we find that individual-level observables explain between 8% and 33% of the urban wage premium, with the individual characteristics explaining a larger portion of the premium since 1980.

As Yankow (2006) and Combes et al. (2008) point out, there may be *unobservable* individual attributes that are correlated with the urban indicator. Thus, we should not try to provide a causal interpretation to these estimates. Unlike prior literature, we do not observe the same individual multiple times, so we cannot include individual-specific fixed effects and therefore rely on a small number of control variables. In section 4.3, we revisit this issue by following a recently proposed method by Altonji and Mansfield

(2018) which allows to control for unobserved covariates by using group-averages of observed variables that we don't have individual-level data for.

4.2 Market Access

As noted above, market access may impact the urban wage premium through potential agglomeration forces. Being located in cities may increase income for workers if it improves firms' access to larger markets and hence increases profits. Market access is a variable that summarizes a locations' access to markets, so controlling for it in our regression framework should remove the portion of the urban wage premium due to being located near other markets.

We summarize our baseline findings in Figure 4 by the line with triangle markers. The urban wage premium coefficient is essentially unchanged from the estimates in Panel B. This suggests that the wage premium due to cities is not being driven by cities' better access to nearby markets and their advantageous location within the transportation network.

4.3 Group Averages

Although we include a small set of individual-level control variables, there are many sources of unobserved covariates that can sort between urban and non-urban areas and lead to differences in average pay. We use the method proposed by Altonji and Mansfield (2018) to attempt to control for these unobserved variables. The proposed method relies on two relationships in how individuals sort. First, group-level averages of variables in urban and non-urban areas reveal information about the differences in various amenities that cause people to sort between urban and non-urban areas.¹⁵ That is, the

¹⁵This can even include the wage premium itself.

averages of covariates in urban and non-urban areas are correlated with average amenity quantities in urban and non-urban areas. Second, individuals sort based on their unobserved covariates because of the *same* set of amenities. That is, the average of unobserved covariates are also correlated with the average amenity quantities.

Hence, controlling for the average of observable covariates should therefore, at least partially, control for averages of unobservable covariates. Altonji and Mansfield (2018) show that under a set of conditions discussed below (and more formally in the authors' work), there exists a *linear* mapping between group-level averages of *observed* covariates and group-level averages of *unobserved* covariates.¹⁶ Therefore including group-level averages linearly in Equation 4 will absorb the portion of the urban wage premium that is due to sorting on unobservable covariates.¹⁷

The result proposed by Altonji and Mansfield (2018) relies on a few assumptions. First, it is assumed that individuals choose locations following a standard logistic choice model with a latent function depending on both observed and unobserved covariates. The second assumption, called the 'rank condition' in their paper, pertains with the set of factors that drive sorting based on the observable group averages. Essentially, this set of factors must contain *all* factors that drive sorting based on unobservable covariates. The intuition for this is that group averages reveal information on only the factors that cause sorting on these covariates, however we need information on all factors in order to control for group-averages of unobservables. The second part of the rank condition is that the number of group-average variables must be larger than the total number of factors driving sorting into urban areas. The intuition for this is that regression will first 'invert' the observed

¹⁶This is true even when there is no correlation between unobservable and observable variables at an individual-level.

¹⁷The group-level averages also will absorb the effects of the amenities that drive sorting, so changes in the wage premium coefficient reflect removal of both sources.

group-level averages to amenities, but this inversion is only possible if the mapping is one-to-one. If there were more amenities than group-average variables, then there would not be a unique inverse and therefore the mapping between observable averages and unobservable averages would break.

The group-level averages we include are ...

The results are presented in Figure 2 by the line with diamond markers

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A. Full Urban Wage Premium Regressions

Full results of our regression estimates in Figure 2 can be found in Table ??.

In Panel A corresponding to the dotted line with diamonds in Figure 2, this set of regressions only have an indicator for living in an Urban Area. In Panel B corresponding to the dotted line with triangle markers in Figure 2, this regression adds controls for age in 5-year bins, education levels, and an indicator for being White. Lastly, Panel C replicates the full regression including controls and the log of market access which is the same as Table ??.

B. MSA-level Market Access

We aggregate our county-level measure of Market Access from Equation 2 to a measure at the urban area. For each year, we start with a crosswalk provided by IPUMS that lists which counties fall (at least partially) inside the boundaries of the metro area. Our metro area boundaries are the 1950 Standard Metropolitan Area for 1940 and 1950, the Metropolitan Statistical Areas for their respective year in 1960-2010, and the Core-Based Statistical Area for 2015.

This list is modified slightly to include MSAs that are recorded in the IPUMS survey data but are not included in that year's crosswalk. The counties that the MSA covers are identified by hand using the shape files from NHGIS for that year or if missing from the previous year.

For a given year, a metro area's market access is the geometric average of the corresponding counties' market access. Then, for our measure of market access for the nonurban portion of a state is the geometric average of all counties that are not within any metro areas. We include robustness checks for weighting the average by county population.

B.1 Robustness Checks

B.1.1 MA Weighted by Population

For some metro areas, there is a central county and more counties at the peripheral, less populated counties. In this case, our simple average may understate the market access in a city, so we check for this by averaging across counties weighted by the counties population share. The results in Figure B.1 show that the shape of the black line with circle markers is very similar to Figure 4.

B.1.2 MA Without Own GDP

Since our measure of market access is increasing in a county's own size, there could be concern that our measure is not only measuring the county's access to the transportation network, but also the size of the county itself. D'Costa and Overman (2014) finds evidence that the urban wage premium indeed is growing in city size. This would mean market access is absorbing some of the premium associated with being in a largely populated area and would be biasing our estimates of the wage premium towards zero. To avoid this, for all counties in an MSA, we construct a measure of market access that removes all counties from the summation that fall within the same MSA:

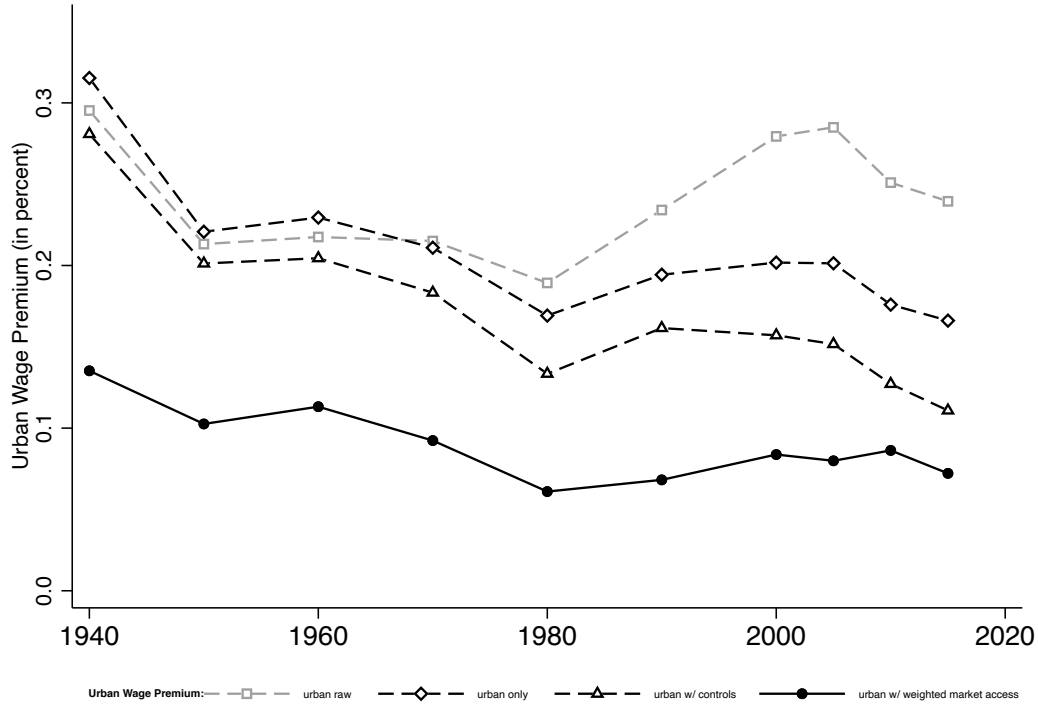
$$\text{MA}_c = \rho \sum_{d \notin \text{MSA}(c)} \tau_{cd}^{-\theta} \text{MA}_d^{-1} Y_d, \quad (5)$$

The results are in Figure B.2.

B.1.3 MSA Selection Over Time

Over decades as places grew in size, more and more places became classified as urban areas and these areas tended to be smaller than long-standing

Figure B.1: Urban Wage Premium 1940-2015 with Weighted Market Access

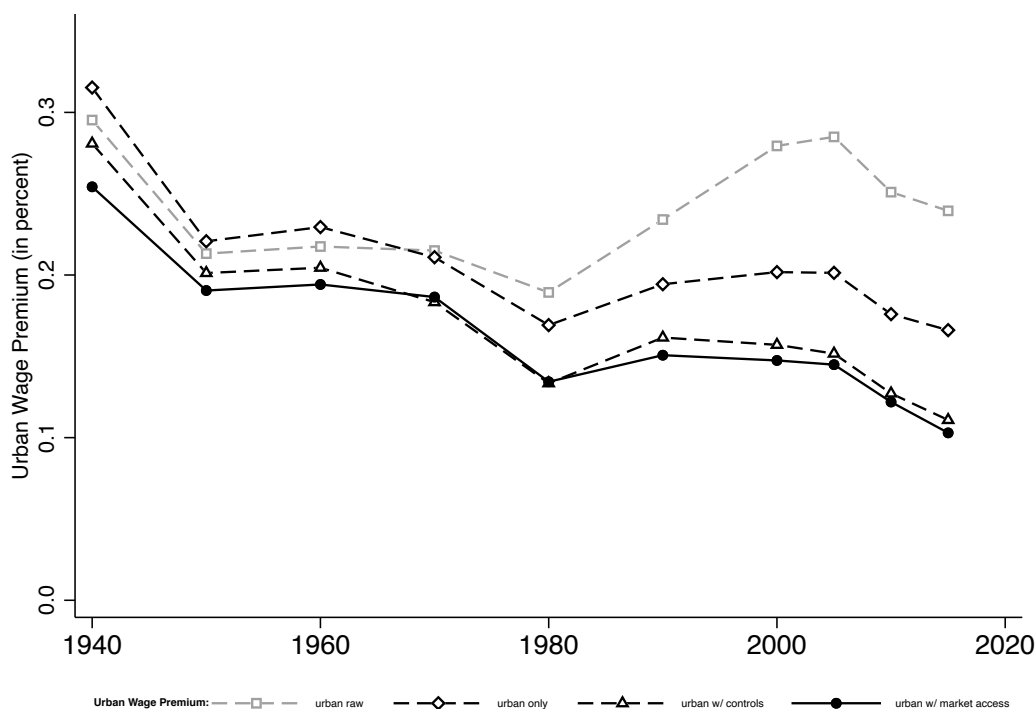


Notes: This figure plots the coefficient estimates for survey-year level estimates of the urban wage premium. In the grey line with square markers, the estimate is found by taking the ratio of the average log weekly wage for urban and nonurban workers. The remaining lines run versions of the regression given by Equation 4. In the line with diamond markers, there are no controls and no measure of market access; the line with triangle markers includes controls for age in 5-year bins, education levels, and an indicator for being White; the line with circle markers includes the same controls and the log of the weighted market access.

MSAs. There could be concern that the observed decline in the urban wage premium as smaller cities are added over time, the observed decline in the urban wage premium is driven by the addition of smaller cities over time. Figure B.3 shows this is not the case.

To create this, we keep all MSAs that exist in the entire sample from 1940 - 2015 (removing those that are added over time and the few that are lost over

Figure B.2: Urban Wage Premium 1940-2015 with Remove Own Market Access

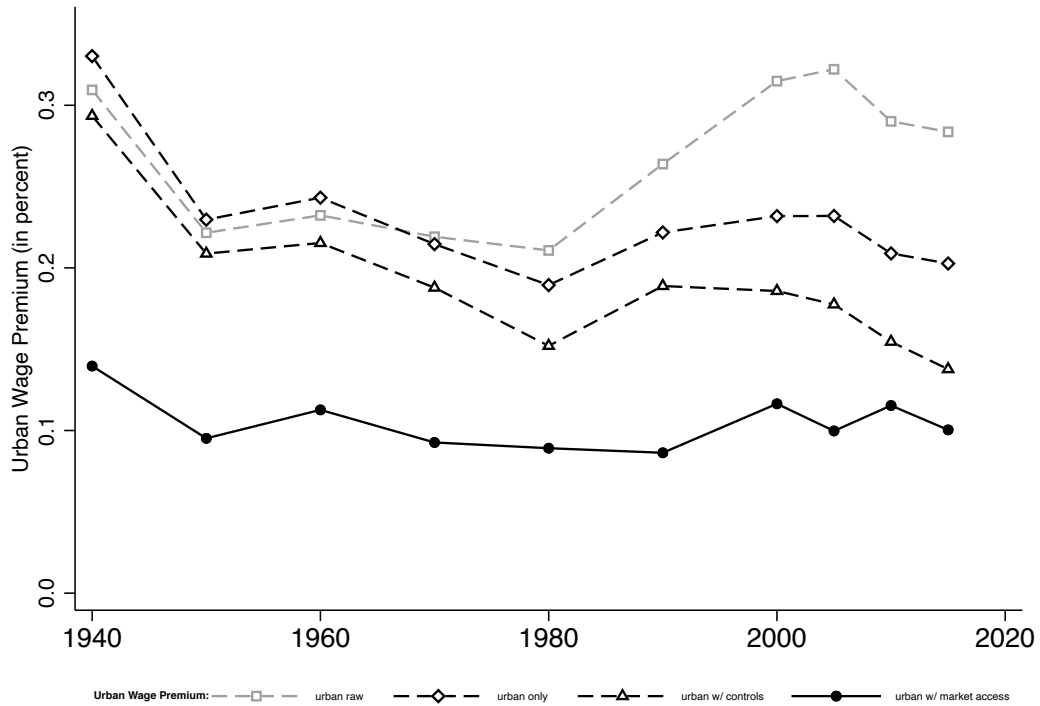


Notes: This figure plots the coefficient estimates for survey-year level estimates of the urban wage premium. In the grey line with square markers, the estimate is found by taking the ratio of the average log weekly wage for urban and nonurban workers. The remaining lines run versions of the regression given by Equation 4. In the line with diamond markers, there are no controls and no measure of market access; the line with triangle markers includes controls for age in 5-year bins, education levels, and an indicator for being White; the line with circle markers includes the same controls and the log of the weighted market access constructed according to Equation 3.

time). Observations are dropped for individuals identified in an MSA that does not exist for the entire sample. There still remains the issue that MSAs tend to grow over time incorporating less densely populated areas over time which we are unable to fix because the survey data does not feature exact

locations of individuals.

Figure B.3: Urban Wage Premium 1940-2015 with Weighted Market Access



Notes: This figure plots the coefficient estimates for survey-year level estimates of the urban wage premium. In the grey line with square markers, the estimate is found by taking the ratio of the average log weekly wage for urban and nonurban workers. The remaining lines run versions of the regression given by Equation 4. In the line with diamond markers, there are no controls and no measure of market access; the line with triangle markers includes controls for age in 5-year bins, education levels, and an indicator for being White; the line with circle markers includes the same controls and the log of the weighted market access.

Table B.1: Road Classifications and Speed Parameters

Class of Road	Speed limit	Speed limit
Interstate	65	70
US Highway (Paved)	45	55
US Highway (Gravel)	35	50
US Highway (Dirt)	30	45
State Highway (Paved)	25	45
State Highway (Gravel)	20	40
State Highway (Dirt)	15	35
Access Road	10	10

Notes: